Data Analysis Plan and Power Justification: Aims 1 and 2

Introduction

This report provides the analysis plan and sample size justification for Aims 1 and 2 of a longitudinal study of innate immune mechanisms of cognitive decline in amnestic mild cognitive impairment (aMCI). Data are collected at baseline and one-year follow-up at the Rocky Mountain Alzheimer's Disease Center (RMADC). Measures include peripheral plasma cytokine/chemokine panels (IL-6, TNF-α, MCP-1, Eotaxin-1), cognitive assessments, structural MRI, and Florbetapir-PET amyloid imaging (global SUVR). Six outcomes will be analyzed separately as one-year change scores: five memory measures (CVLT 20-minute delayed recall, CVLT recognition d-prime, Benson Figure 15-minute delayed recall, MST pattern separation proportion correct, and Story Recall 1-week delayed recall) and AD-signature mean cortical thickness (FreeSurfer 5.1). The target enrollment is 137 aMCI and 55 healthy controls (HC; N = 192), yielding N = 175 after 10% attrition. A preliminary dataset (n = 30) was provided to inform effect size estimates.

The scientific hypotheses translate to the following statistical tests. **H1a:** A significant negative regression coefficient for baseline cytokine/chemokine level predicting one-year decline in each outcome. **H1b:** A significant negative regression coefficient for change in cytokine/chemokine level predicting the corresponding decline. **H2:** A significant cytokine $\times$ amyloid SUVR interaction coefficient, indicating steeper decline when both elevated inflammation and significant amyloid burden are present.

Methods

Data Handling. Distributional properties of cytokine/chemokine variables will be examined via Q-Q plots and Shapiro-Wilk tests; log-transformation will be applied only if warranted (preliminary tests suggest normality; all $p > 0.09$). Values below the detection limit will be substituted at half that limit; outliers exceeding 4 SD will be flagged. Primary analyses will use available cases under a missing-at-random assumption; multiple imputation will serve as a sensitivity analysis if dropout is associated with baseline characteristics.

Aim 1. Multivariable linear regression will model each one-year change score (follow-up minus baseline) as the outcome, with the baseline outcome value as a covariate to adjust for regression to the mean. For H1a, the predictor is baseline cytokine level; for H1b, the cytokine change score. Separate models will be fit for each of four markers across each of six outcomes: CVLT delayed recall, CVLT recognition d-prime, Benson Figure delayed recall, MST pattern separation, Story Recall 1-week delay, and AD-signature cortical thickness. Markers will not serve as simultaneous predictors due to extreme inter-marker correlation (preliminary IL-6/MCP-1: $r = 0.93$). Hypotheses are tested via two-sided t-tests on the cytokine regression coefficient. All models adjust for age, sex, APOE 4, BMI, hypercholesterolemia, NSAID use, immune-related conditions, and diagnostic group (aMCI vs. HC). Both groups are combined to maximize cytokine and outcome range; effect modification by group will be examined secondarily. Collinearity will be assessed via VIF (>5 triggers investigation) and the change-in-estimate criterion (>10% change warrants retention). Total cortical thickness will be included in the cortical thickness model to isolate AD-signature effects. Table A1 (Appendix) summarizes all variables.

Aim 2. Aim 2 extends Aim 1 by adding baseline amyloid SUVR and its product interaction with each cytokine. The primary test of H2 is the interaction coefficient, evaluated via a t-test (equivalently, an F-test with 1 numerator df for the R^2 increment). Both main effects and the same covariates from Aim 1 are included. Secondarily, amyloid will be dichotomized (SUVR $\geq$ 1.10); significant interactions will be probed via simple slopes at ± 1 SD of the cytokine, yielding a 2×2 pattern. The hypothesis predicts the steepest decline in the amyloid-positive/high-inflammation cell. Sensitivity analyses stratified by APOE 4 will evaluate whether the interaction differs by genotype.

Multiple Comparisons. Within each of the six outcomes, false discovery rate (FDR;

Benjamini-Hochberg, $q < 0.05$) will be applied across the four inflammatory markers. FDR is preferred over Bonferroni because the markers are substantially correlated (preliminary r up to 0.93), making family-wise error control overly conservative. Bonferroni ($\alpha/4 = 0.0125$) will be a sensitivity analysis. The six outcomes will not be corrected across one another; concordant patterns across related outcomes will be interpreted as strengthening overall evidence.

Power Analysis and Sample Size Justification

We estimated power for N = 175 (125 aMCI + 50 HC, after 10% attrition from 192 enrolled). Effect sizes were derived from Pearson correlations in the preliminary dataset ($n = 30$), ranging from $|r| = 0.26$ (IL-6 vs. memory change) to $|r| = 0.69$ (MCP-1 vs. cortical thickness change). All tests were two-sided; power is presented at $\alpha = 0.05$ and $\alpha = 0.0125$ (Bonferroni $\alpha/4$), with the effective FDR threshold falling between these values.

Aim 1. Power was estimated using `pwr.r.test()` (R `pwr` package), approximating the adjusted regression as a bivariate correlation — conservative since covariate adjustment generally increases precision. The minimum detectable $|r|$ at 80% power was 0.21 ($\alpha = 0.05$) and 0.249 ($\alpha = 0.0125$). The weakest preliminary signal (IL-6/memory, $|r| = 0.26$) achieved 94% and 84% power; all stronger associations exceeded 96% at either level (Table 1, Figure 1A).

Aim 2. Power for the interaction R^2 increment was estimated using `pwr.f2.test()` (R `pwr` package; 10 predictors, numerator df = 1, denominator df = 164). The minimum detectable f^2 at 80% power was 0.048 ($\alpha = 0.05$) and 0.068 ($\alpha = 0.0125$), both small-to-medium by Cohen's benchmarks. At $f^2 = 0.05$, power was 82% ($\alpha = 0.05$); at $f^2 = 0.08$, power reached 87% at $\alpha = 0.0125$ (Table 2, Figure 1B). Assuming $R^2_{\text{full}} \approx 0.30$, the detectable interaction corresponds to $\Delta R^2 \approx 3.4\%$. The proposed N is justified by RMADC enrollment feasibility.

Tables and Figures

Table 1: Aim 1 Power: Bivariate Correlation Test (N = 175, two-sided). Computed using pwr.r.test(), R pwr package. Preliminary |r| ranged 0.26 to 0.69.

|r|	Power (a = 0.05)	Power (a = 0.0125)
0.10	0.261	0.119
0.15	0.511	0.304
0.20	0.760	0.565
0.25	0.919	0.804
0.30	0.983	0.942
0.35	0.998	0.989
0.40	1.000	0.999
0.45	1.000	1.000
0.50	1.000	1.000

Table 2: Aim 2 Power: Interaction F-test (N = 175, u = 1, v = 164). Computed using pwr.f2.test(), R pwr package. dR2 approximated assuming R2_full = 0.30. Cohen benchmarks: small = 0.02, medium = 0.15.

f2	dR2	Power (a = 0.05)	Power (a = 0.0125)
0.02	0.014	0.441	0.244
0.03	0.021	0.602	0.387
0.04	0.028	0.726	0.522

f2	dR2	Power (a = 0.05)	Power (a = 0.0125)
0.05	0.035	0.817	0.639
0.06	0.042	0.880	0.735
0.08	0.056	0.952	0.867
0.10	0.070	0.982	0.938
0.15	0.105	0.999	0.993

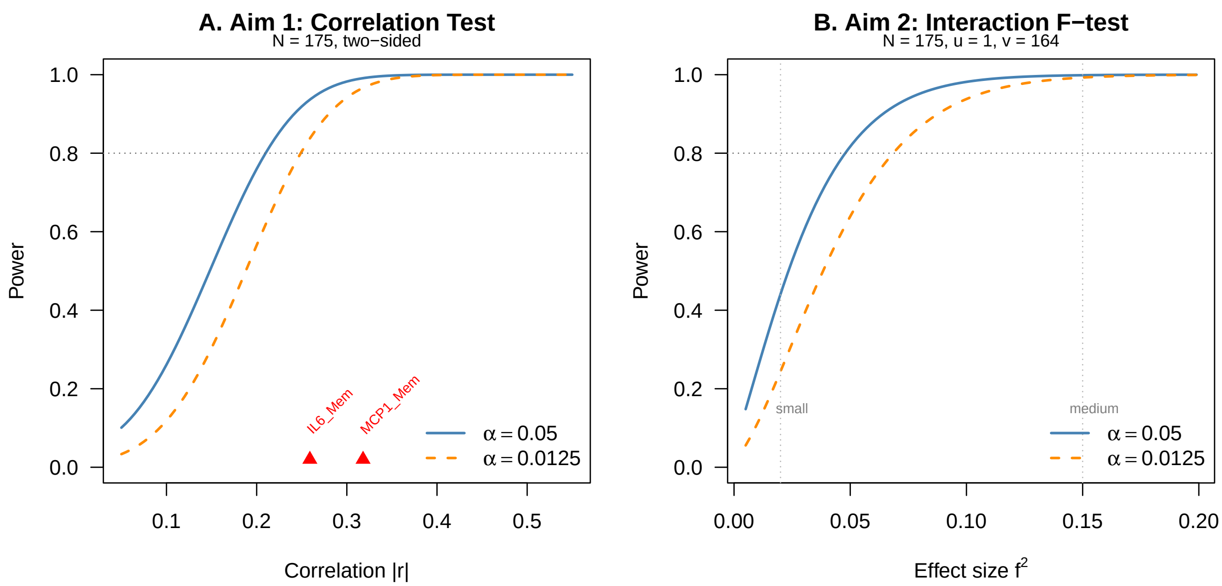

Figure 1: Power curves for Aim 1 (bivariate correlation) and Aim 2 (interaction F-test) at N = 175. Solid blue: alpha = 0.05; dashed orange: alpha = 0.0125 (Bonferroni alpha/4). Red triangles in Panel A mark preliminary data correlations.

Appendix

Table A1. Variable Summary by Role

Role	Variable	Type	Measurement / Coding
Outcome 1	Change in CVLT delayed recall	Continuous	20-min delay, total correct; follow-up - baseline
Outcome 2	Change in CVLT recognition	Continuous	Recognition d-prime; follow-up - baseline
Outcome 3	Change in Benson Figure recall	Continuous	15-min delay, total correct; follow-up - baseline
Outcome 4	Change in MST pattern separation	Continuous	Proportion correct; follow-up - baseline
Outcome 5	Change in Story Recall	Continuous	1-week delay, total correct; follow-up - baseline
Outcome 6	Change in AD-signature cortical thickness	Continuous	FreeSurfer 5.1 AD-signature ROI; follow-up - baseline
Primary Predictor (Aim 1)	Plasma cytokines/chemokines (IL-6, TNF-a, MCP-1, Eotaxin-1)	Continuous	pg/mL; transform if warranted; H1a: baseline; H1b: change
Primary Predictor (Aim 2)	Amyloid SUVR x cytokine interaction	Continuous; Binary	Primary: SUVR x cytokine product; Secondary: amyloid-pos/neg (SUVR >= 1.10) x cytokine

Role	Variable	Type	Measurement / Coding
Covariates (All Models)	Age, sex, APOE e4, BMI, hypercholesterolemia, NSAID use, immune conditions, diagnostic group, baseline outcome	Continuous; Binary	Pre-specified; baseline outcome adjusts for regression to the mean

Reproducible Research

Data location: `Project_2/DataRaw/PrelimData.csv`

Code location: `Project_2/Code/Project2_PowerAnalysis.R` (standalone power analysis); `Project_2/Code/Project2_Report.Rmd` (this report)

Results location: `Project_2/Results/` (power tables as CSV; power curve figure as PDF)

GitHub repository: https://github.com/Charl3456/BIOS6624.git

Software: R version 4.4.1; `pwr` package (functions: `pwr.r.test()`, `pwr.f2.test()`)